BSDS (2025 Semester II): Statistics II

Midsem: Part A

NAME :	**ROLL :**

Time: 60 minutes **Total marks:** 44

All the rough works should be done in the space provided in the question paper.

I. *Answer all questions. Each question has one or more correct options. Identify all of them.*
$6 \times 4 = 24$

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1. Let $X_1, \ldots, X_n$ be i.i.d. samples from $F_\theta \equiv \texttt{Uniform}(0, \theta)$. Indicate which of the following statements are correct?

   A. $(\theta - \bar{X}_n)$ is an unbiased estimator of $\theta/2$. ☐

   B. $\bar{X}_n$ is an unbiased estimator of $\theta/2$. ☐

   C. Sample variance is the MoM estimator of the population variance. ☐

   D. $\bar{X}_n^2/3$ is the MoM estimator of the population variance. ☐

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2. Let T_1, T_2 and T_3 be three estimators of $g(\theta)$.

   A. If T_1 is unbiased for $g(\theta)$, then $\log T_1$ is unbiased for $\log g(\theta)$. ☐

   B. If T_3 is an MLE for $g(\theta)$, then $\log T_3$ is an MLE for $\log g(\theta)$. ☐

   C. If both T_1 and T_2 are unbiased for $g(\theta)$ then $(2T_1 + T_2)/3$ is also unbiased for $g(\theta)$. ☐

   D. Among all the three estimators of $g(\theta)$, the one with minimum variance is the *best*. ☐

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3. Suppose you need to infer the population variance σ^2, towards which you consider a random sample of size n, say, $X_1, \ldots, X_n$. Indicate which of the following statements are correct.

   A. Let T_n be an estimator of σ^2 with expectation $(n+2)\sigma^2/(n+1)$ and variance $\sigma^4/(n-1)^2$, and S_n be an unbiased estimator of σ^2 with variance $2\sigma^4/n^2$. Then S_n is a better estimator of σ^2 than T_n. ☐

   B. Consider the estimator T_n described in part A. As n increases, the means of the sampling distributions of T_n necessarily become closer to σ^2. ☐

   C. Consider the estimator T_n described in part A. As n increases, the realizations of T_n necessarily become closer to σ^2. ☐

   D. Suppose you take another sample of same size, say, $Y_1, \ldots, Y_n$, then the realizations of $\bar{Y}_n$ will necessarily be the same as the realization of $\bar{X}_n$. ☐

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4. A person reaches his nearest bus-stop at 10 AM and waits for a particular bus. We are interested in studying the mean waiting time until the first bus arrives. Let X denote the waiting time (in minutes) for the person to catch the first bus. Determine which of the following distributions are appropriate for the model.

   A. X follows a `Geometric`(θ) distribution with probability mass function

   $$f_\theta(x) = \theta(1-\theta)^{x-1}, \quad x = 1, 2, \ldots, \quad 0 < \theta < 1.$$ ☐

   B. X follows a `Geometric`(θ) distribution. with probability mass function

   $$f_\theta(x) = \theta(1-\theta)^{x}, \quad x = 0, 1, \ldots, \quad 0 < \theta < 1.$$ ☐

   C. X follows an `Binomial`(N, θ) distribution for some $N \in \mathbb{N}$. ☐

   D. X follows a `Poisson`(θ) distribution. ☐

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5. Let X be a random variable with the following distribution:

x	-3	-2	-1	0	1	2	3
$P(X = x)$	0.05	0.10	0.15	0.4	0.15	0.10	0.05

Consider the random variable $Y = \log |X|$. Indicate which of the following statements are true.

A. $E(X^{21}) = 0$. ☐

B. $E(Y^{21}) = -\infty$. ☐

C. $P(Y = 0) = 0.15$. ☐

D. $P(Y = 0) = 0.40$. ☐

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6. Let $X_1, \ldots, X_{2n+1}$, $n \in \mathbb{N}$, be i.i.d. samples from $N(0, 1)$ distribution. Indicate which of the following statements are correct:

A. $\dfrac{X_1^2 + X_3^2 + \ldots + X_{2n+1}^2}{X_2^2 + X_4^2 + \ldots + X_{2n}^2} \sim F_{n,n}$. ☐

B. $2^{-1}(X_1^2 + X_2^2 + \ldots + X_{2n}^2) \sim \chi_n^2$. ☐

C. $X_{2n+1}/\sqrt{X_1^2 + \ldots + X_{2n}^2} \sim t_{2n}$. ☐

D. $\dfrac{2nX_1^2}{X_1^2 + X_2^2 + \ldots + X_{2n}^2} \sim F_{1,2n}$. ☐

II. *Write the answers to the following questions in one sentence.* $5 \times 4 = 20$

1. Let $X_1, \ldots, X_n$ be i.i.d. random variables from the inverse Gaussian distribution with parameters $\boldsymbol{\theta} = (\mu, \lambda)$ and common pdf

$$f_{\boldsymbol{\theta}}(x) = \sqrt{\frac{\lambda}{2\pi x^3}} \exp\left\{-\frac{\lambda(x-\mu)^2}{2\mu^2 x}\right\}, \quad x > 0,\ \mu > 0,\ \lambda > 0.$$

Then a complete and sufficient statistic for $\boldsymbol{\theta}$ is:

2. Let $X_1, \ldots, X_n$ be i.i.d. random variables from a distribution with the following p.d.f.

$$f(x) = \begin{cases} (2x^3)^{-1} & \text{for} \quad x \geq 1/2, \\ 0 & \text{otherwise.} \end{cases}$$

Then the pdf of $Y = X_{(1)}$ is:

3. Let $X \sim \texttt{Uniform}(0, 10.5)$. Let $Y = \lfloor X \rfloor$, where $\lfloor a \rfloor$ is the greatest integer not exceeding a. Then $E(Y)$ is *(write in mixed fraction)*:

4. Let $X_1, \ldots, X_n$ be i.i.d. random variables from the discrete uniform distribution supported on $\{1, \ldots, N\}$, $N \in \mathbb{N}$. Then the MLE of N is :

5. Let S_n be an unbiased estimator of $\sigma^2 \in (0, \infty)$ with variance $8\sigma^4/\{5(n-1)^2\}$. The minimal sample size n such that the ratio S_n/σ^2 lies between $[1/2, 3/2]$ with probability at least 0.9 is :

Rough work